

Model for “Th”

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General Condition

- Basis type: jml
- SAMB selection:
 - Type: [Q, G]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_g , A_u , E_g , E_u , T_g , T_u]
 - Spin (s): [0, 1]
- Atomic selection:
 - Type: [Q, G, M, T]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_g , A_u , E_g , E_u , T_g , T_u]
 - Spin (s): [0, 1]
- Site-cluster selection:
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_g , A_u , E_g , E_u , T_g , T_u]
- Bond-cluster selection:
 - Type: [Q, G, M, T]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_g , A_u , E_g , E_u , T_g , T_u]
- Max. neighbor: 10
- Search cell range: (-2, 3), (-2, 3), (-2, 3)
- Toroidal priority: **false**

Group and Unit Cell

- Group: PG No. 29 T_h $m\bar{3}$ [cubic]
- Unit cell:
 - $a = 1.00000$, $b = 1.00000$, $c = 1.00000$, $\alpha = 90.0$, $\beta = 90.0$, $\gamma = 90.0$
- Lattice vectors (conventional cell):
 - $\mathbf{a}_1 = [1.00000, 0.00000, 0.00000]$
 - $\mathbf{a}_2 = [0.00000, 1.00000, 0.00000]$
 - $\mathbf{a}_3 = [0.00000, 0.00000, 1.00000]$

Symmetry Operation

Table 1: Symmetry operation

#	SO	#	SO	#	SO	#	SO	#	SO
1	1	2	2 ₀₀₁	3	2 ₀₁₀	4	2 ₁₀₀	5	3 ⁺ ₁₁₁
6	3 ⁺ ₋₁₁₋₁	7	3 ⁺ ₁₋₁₋₁	8	3 ⁺ ₋₁₋₁₁	9	3 ⁻ ₁₁₁	10	3 ⁻ ₁₋₁₋₁
11	3 ⁻ ₋₁₋₁₁	12	3 ⁻ ₋₁₁₋₁	13	-1	14	m ₀₀₁	15	m ₀₁₀
16	m ₁₀₀	17	-3 ⁺ ₁₁₁	18	-3 ⁺ ₋₁₁₋₁	19	-3 ⁺ ₁₋₁₋₁	20	-3 ⁺ ₋₁₋₁₁
21	-3 ⁻ ₁₁₁	22	-3 ⁻ ₁₋₁₋₁	23	-3 ⁻ ₋₁₋₁₁	24	-3 ⁻ ₋₁₁₋₁		

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{Q}_0(A_g)$	A_g	0	Q, T	-	-	1
2	$\mathbb{G}_3(A_g)$	A_g	3	G, M	-	-	$\sqrt{15}xyz$
3	$\mathbb{Q}_4(A_g)$	A_g	4	Q, T	-	-	$\frac{\sqrt{21}(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4)}{6}$
4	$\mathbb{G}_0(A_u)$	A_u	0	G, M	-	-	1
5	$\mathbb{Q}_3(A_u)$	A_u	3	Q, T	-	-	$\sqrt{15}xyz$
6	$\mathbb{G}_4(A_u)$	A_u	4	G, M	-	-	$\frac{\sqrt{21}(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4)}{6}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
7	$\mathbb{Q}_{2,1}(E_g)$	E_g	2	Q, T	-	1	$\frac{-x^2-y^2+2z^2}{2}$
8	$\mathbb{Q}_{2,2}(E_g)$					2	$\frac{\sqrt{3}(x^2-y^2)}{2}$
9	$\mathbb{Q}_{4,1}(E_g)$	E_g	4	Q, T	-	1	$\frac{\sqrt{15}(-x^4+12x^2y^2-6x^2z^2-y^4-6y^2z^2+2z^4)}{12}$
10	$\mathbb{Q}_{4,2}(E_g)$					2	$\frac{\sqrt{5}(x^4-6x^2z^2-y^4+6y^2z^2)}{4}$
11	$\mathbb{G}_{2,1}(E_u)$	E_u	2	G, M	-	1	$\frac{-x^2-y^2+2z^2}{2}$
12	$\mathbb{G}_{2,2}(E_u)$					2	$\frac{\sqrt{3}(x^2-y^2)}{2}$
13	$\mathbb{G}_{4,1}(E_u)$	E_u	4	G, M	-	1	$\frac{\sqrt{15}(-x^4+12x^2y^2-6x^2z^2-y^4-6y^2z^2+2z^4)}{12}$
14	$\mathbb{G}_{4,2}(E_u)$					2	$\frac{\sqrt{5}(x^4-6x^2z^2-y^4+6y^2z^2)}{4}$
15	$\mathbb{Q}_{5,1}(E_u)$	E_u	5	Q, T	-	1	$\frac{3\sqrt{35}xyz(x^2-y^2)}{2}$
16	$\mathbb{Q}_{5,2}(E_u)$					2	$\frac{\sqrt{105}xyz(x^2+y^2-2z^2)}{2}$
17	$\mathbb{G}_{1,1}(T_g)$	T_g	1	G, M	-	1	x
18	$\mathbb{G}_{1,2}(T_g)$					2	y
19	$\mathbb{G}_{1,3}(T_g)$					3	z
20	$\mathbb{Q}_{2,1}(T_g)$	T_g	2	Q, T	-	1	$\sqrt{3}yz$
21	$\mathbb{Q}_{2,2}(T_g)$					2	$\sqrt{3}xz$
22	$\mathbb{Q}_{2,3}(T_g)$					3	$\sqrt{3}xy$
23	$\mathbb{G}_{3,1}(T_g, 1)$	T_g	3	G, M	1	1	$\frac{x(2x^2-3y^2-3z^2)}{2}$
24	$\mathbb{G}_{3,2}(T_g, 1)$					2	$\frac{y(-3x^2+2y^2-3z^2)}{2}$
25	$\mathbb{G}_{3,3}(T_g, 1)$					3	$\frac{z(-3x^2-3y^2+2z^2)}{2}$
26	$\mathbb{G}_{3,1}(T_g, 2)$	T_g	3	G, M	2	1	$\frac{\sqrt{15}x(y^2-z^2)}{2}$
27	$\mathbb{G}_{3,2}(T_g, 2)$					2	$\frac{\sqrt{15}y(-x^2+z^2)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
28	$\mathbb{G}_{3,3}(T_g, 2)$					3	$\frac{\sqrt{15}z(x^2-y^2)}{2}$
29	$\mathbb{Q}_{4,1}(T_g, 1)$	T_g	4	Q, T	1	1	$\frac{\sqrt{35}yz(y^2-z^2)}{2}$
30	$\mathbb{Q}_{4,2}(T_g, 1)$					2	$\frac{\sqrt{35}xz(-x^2+z^2)}{2}$
31	$\mathbb{Q}_{4,3}(T_g, 1)$					3	$\frac{\sqrt{35}xy(x^2-y^2)}{2}$
32	$\mathbb{Q}_{4,1}(T_g, 2)$	T_g	4	Q, T	2	1	$\frac{\sqrt{5}yz(6x^2-y^2-z^2)}{2}$
33	$\mathbb{Q}_{4,2}(T_g, 2)$					2	$\frac{\sqrt{5}xz(-x^2+6y^2-z^2)}{2}$
34	$\mathbb{Q}_{4,3}(T_g, 2)$					3	$\frac{\sqrt{5}xy(-x^2-y^2+6z^2)}{2}$
35	$\mathbb{Q}_{1,1}(T_u)$	T_u	1	Q, T	-	1	x
36	$\mathbb{Q}_{1,2}(T_u)$					2	y
37	$\mathbb{Q}_{1,3}(T_u)$					3	z
38	$\mathbb{G}_{2,1}(T_u)$	T_u	2	G, M	-	1	$\sqrt{3}yz$
39	$\mathbb{G}_{2,2}(T_u)$					2	$\sqrt{3}xz$
40	$\mathbb{G}_{2,3}(T_u)$					3	$\sqrt{3}xy$
41	$\mathbb{Q}_{3,1}(T_u, 1)$	T_u	3	Q, T	1	1	$\frac{x(2x^2-3y^2-3z^2)}{2}$
42	$\mathbb{Q}_{3,2}(T_u, 1)$					2	$\frac{y(-3x^2+2y^2-3z^2)}{2}$
43	$\mathbb{Q}_{3,3}(T_u, 1)$					3	$\frac{z(-3x^2-3y^2+2z^2)}{2}$
44	$\mathbb{Q}_{3,1}(T_u, 2)$	T_u	3	Q, T	2	1	$\frac{\sqrt{15}x(y^2-z^2)}{2}$
45	$\mathbb{Q}_{3,2}(T_u, 2)$					2	$\frac{\sqrt{15}y(-x^2+z^2)}{2}$
46	$\mathbb{Q}_{3,3}(T_u, 2)$					3	$\frac{\sqrt{15}z(x^2-y^2)}{2}$
47	$\mathbb{G}_{4,1}(T_u, 1)$	T_u	4	G, M	1	1	$\frac{\sqrt{35}yz(y^2-z^2)}{2}$
48	$\mathbb{G}_{4,2}(T_u, 1)$					2	$\frac{\sqrt{35}xz(-x^2+z^2)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
49	$\mathbb{G}_{4,3}(T_u, 1)$					3	$\frac{\sqrt{35}xy(x^2-y^2)}{2}$
50	$\mathbb{G}_{4,1}(T_u, 2)$	T_u	4	G, M	2	1	$\frac{\sqrt{5}yz(6x^2-y^2-z^2)}{2}$
51	$\mathbb{G}_{4,2}(T_u, 2)$					2	$\frac{\sqrt{5}xz(-x^2+6y^2-z^2)}{2}$
52	$\mathbb{G}_{4,3}(T_u, 2)$					3	$\frac{\sqrt{5}xy(-x^2-y^2+6z^2)}{2}$
53	$\mathbb{Q}_{5,1}(T_u, 3)$	T_u	5	Q, T	3	1	$\frac{\sqrt{105}x(2x^2y^2-2x^2z^2-y^4+z^4)}{4}$
54	$\mathbb{Q}_{5,2}(T_u, 3)$					2	$\frac{\sqrt{105}y(x^4-2x^2y^2+2y^2z^2-z^4)}{4}$
55	$\mathbb{Q}_{5,3}(T_u, 3)$					3	$\frac{\sqrt{105}z(-x^4+2x^2z^2+y^4-2y^2z^2)}{4}$

— Basis in full matrix —

Table 3: dimension = 48

#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
0	$ \frac{1}{2}, \frac{1}{2}; p\rangle @A(1)$	1	$ \frac{1}{2}, -\frac{1}{2}; p\rangle @A(1)$	2	$ \frac{3}{2}, \frac{3}{2}; p\rangle @A(1)$	3	$ \frac{3}{2}, \frac{1}{2}; p\rangle @A(1)$	4	$ \frac{3}{2}, -\frac{1}{2}; p\rangle @A(1)$
5	$ \frac{3}{2}, -\frac{3}{2}; p\rangle @A(1)$	6	$ \frac{1}{2}, \frac{1}{2}; p\rangle @A(2)$	7	$ \frac{1}{2}, -\frac{1}{2}; p\rangle @A(2)$	8	$ \frac{3}{2}, \frac{3}{2}; p\rangle @A(2)$	9	$ \frac{3}{2}, \frac{1}{2}; p\rangle @A(2)$
10	$ \frac{3}{2}, -\frac{1}{2}; p\rangle @A(2)$	11	$ \frac{3}{2}, -\frac{3}{2}; p\rangle @A(2)$	12	$ \frac{1}{2}, \frac{1}{2}; p\rangle @A(3)$	13	$ \frac{1}{2}, -\frac{1}{2}; p\rangle @A(3)$	14	$ \frac{3}{2}, \frac{3}{2}; p\rangle @A(3)$
15	$ \frac{3}{2}, \frac{1}{2}; p\rangle @A(3)$	16	$ \frac{3}{2}, -\frac{1}{2}; p\rangle @A(3)$	17	$ \frac{3}{2}, -\frac{3}{2}; p\rangle @A(3)$	18	$ \frac{1}{2}, \frac{1}{2}; p\rangle @A(4)$	19	$ \frac{1}{2}, -\frac{1}{2}; p\rangle @A(4)$
20	$ \frac{3}{2}, \frac{3}{2}; p\rangle @A(4)$	21	$ \frac{3}{2}, \frac{1}{2}; p\rangle @A(4)$	22	$ \frac{3}{2}, -\frac{1}{2}; p\rangle @A(4)$	23	$ \frac{3}{2}, -\frac{3}{2}; p\rangle @A(4)$	24	$ \frac{1}{2}, \frac{1}{2}; p\rangle @A(5)$
25	$ \frac{1}{2}, -\frac{1}{2}; p\rangle @A(5)$	26	$ \frac{3}{2}, \frac{3}{2}; p\rangle @A(5)$	27	$ \frac{3}{2}, \frac{1}{2}; p\rangle @A(5)$	28	$ \frac{3}{2}, -\frac{1}{2}; p\rangle @A(5)$	29	$ \frac{3}{2}, -\frac{3}{2}; p\rangle @A(5)$
30	$ \frac{1}{2}, \frac{1}{2}; p\rangle @A(6)$	31	$ \frac{1}{2}, -\frac{1}{2}; p\rangle @A(6)$	32	$ \frac{3}{2}, \frac{3}{2}; p\rangle @A(6)$	33	$ \frac{3}{2}, \frac{1}{2}; p\rangle @A(6)$	34	$ \frac{3}{2}, -\frac{1}{2}; p\rangle @A(6)$

continued ...

Table 3

#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
35	$ \frac{3}{2}, -\frac{3}{2}; p\rangle @A(6)$	36	$ \frac{1}{2}, \frac{1}{2}; p\rangle @A(7)$	37	$ \frac{1}{2}, -\frac{1}{2}; p\rangle @A(7)$	38	$ \frac{3}{2}, \frac{3}{2}; p\rangle @A(7)$	39	$ \frac{3}{2}, \frac{1}{2}; p\rangle @A(7)$
40	$ \frac{3}{2}, -\frac{1}{2}; p\rangle @A(7)$	41	$ \frac{3}{2}, -\frac{3}{2}; p\rangle @A(7)$	42	$ \frac{1}{2}, \frac{1}{2}; p\rangle @A(8)$	43	$ \frac{1}{2}, -\frac{1}{2}; p\rangle @A(8)$	44	$ \frac{3}{2}, \frac{3}{2}; p\rangle @A(8)$
45	$ \frac{3}{2}, \frac{1}{2}; p\rangle @A(8)$	46	$ \frac{3}{2}, -\frac{1}{2}; p\rangle @A(8)$	47	$ \frac{3}{2}, -\frac{3}{2}; p\rangle @A(8)$				

SAMB: 552 (all 552)

• A : 'A' site-cluster

* bra: $\langle \frac{1}{2}, \frac{1}{2}; p |, \langle \frac{1}{2}, -\frac{1}{2}; p |, \langle \frac{3}{2}, \frac{3}{2}; p |, \langle \frac{3}{2}, \frac{1}{2}; p |, \langle \frac{3}{2}, -\frac{1}{2}; p |, \langle \frac{3}{2}, -\frac{3}{2}; p |$

* ket: $|\frac{1}{2}, \frac{1}{2}; p\rangle, |\frac{1}{2}, -\frac{1}{2}; p\rangle, |\frac{3}{2}, \frac{3}{2}; p\rangle, |\frac{3}{2}, \frac{1}{2}; p\rangle, |\frac{3}{2}, -\frac{1}{2}; p\rangle, |\frac{3}{2}, -\frac{3}{2}; p\rangle$

* wyckoff: **8b**

$$\boxed{\text{z1}} \quad \mathbb{Q}_0^{(c)}(A_g, a) = \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_0^{(s)}(A_g)$$

$$\boxed{\text{z2}} \quad \mathbb{Q}_0^{(c)}(A_g, b) = \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(s)}(T_g) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(s)}(T_g) + \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(s)}(T_g) \right)}{3}$$

$$\boxed{\text{z3}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_g) = \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(s)}(T_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(s)}(T_g) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(s)}(T_g) \right)}{3}$$

$$\boxed{\text{z4}} \quad \mathbb{Q}_0^{(1,1;c)}(A_g) = \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_0^{(s)}(A_g)$$

$$\boxed{\text{z5}} \quad \mathbb{G}_3^{(1,0;c)}(A_g) = \frac{\sqrt{3} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(s)}(T_g) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{2,2}^{(s)}(T_g) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{2,3}^{(s)}(T_g) \right)}{3}$$

$$\boxed{\text{z27}} \quad \mathbb{Q}_3^{(c)}(A_u, a) = \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_3^{(s)}(A_u)$$

$$\boxed{\text{z28}} \quad \mathbb{Q}_3^{(c)}(A_u, b) = \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{1,1}^{(s)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{1,2}^{(s)}(T_u) + \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{1,3}^{(s)}(T_u) \right)}{3}$$

$$\boxed{\text{z29}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,1}^{(s)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,2}^{(s)}(T_u) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,3}^{(s)}(T_u) \right)}{3}$$

$$\boxed{\text{z30}} \quad \mathbb{Q}_3^{(1,1;c)}(A_u) = \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_3^{(s)}(A_u)$$

$$\boxed{\text{z31}} \quad \mathbb{G}_0^{(1,0;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{1,1}^{(s)}(T_u) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{1,2}^{(s)}(T_u) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{1,3}^{(s)}(T_u) \right)}{3}$$

$$\boxed{\text{z47}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z48}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z49}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, b) = \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(s)}(T_g) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(s)}(T_g) - 2 \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(s)}(T_g) \right)}{6}$$

$$\boxed{\text{z50}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, b) = \frac{-\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(s)}(T_g) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(s)}(T_g)}{2}$$

$$\boxed{\text{z51}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z52}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z53}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) = \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(s)}(T_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(s)}(T_g) - 2 \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(s)}(T_g) \right)}{6}$$

$$\boxed{\text{z54}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) = \frac{-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(s)}(T_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(s)}(T_g)}{2}$$

$$\boxed{\text{z55}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(s)}(T_g) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{2,2}^{(s)}(T_g)}{2}$$

$$\boxed{\text{z56}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g) = \frac{\sqrt{3} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(s)}(T_g) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{2,2}^{(s)}(T_g) - 2 \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{2,3}^{(s)}(T_g) \right)}{6}$$

$$\boxed{\text{z99}} \quad \mathbb{Q}_{5,1}^{(c)}(E_u) = \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_3^{(s)}(A_u)}{2}$$

$$\boxed{\text{z100}} \quad \mathbb{Q}_{5,2}^{(c)}(E_u) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_3^{(s)}(A_u)}{2}$$

$$\boxed{\text{z101}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_3^{(s)}(A_u)}{2}$$

$$\boxed{\text{z102}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_3^{(s)}(A_u)}{2}$$

$$\boxed{\text{z103}} \quad \mathbb{G}_{2,1}^{(c)}(E_u) = \frac{-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u)}{2}$$

$$\boxed{\text{z104}} \quad \mathbb{G}_{2,2}^{(c)}(E_u) = \frac{\sqrt{3}\left(-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u) - \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u) + 2\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(s)}(T_u)\right)}{6}$$

$$\boxed{\text{z105}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = \frac{-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u)}{2}$$

$$\boxed{\text{z106}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{3}\left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u) + 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(s)}(T_u)\right)}{6}$$

$$\boxed{\text{z107}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u) = \frac{\sqrt{3}\left(-\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u) + 2\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{1,3}^{(s)}(T_u)\right)}{6}$$

$$\boxed{\text{z108}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u)}{2}$$

$$\boxed{\text{z139}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(s)}(T_g)}{3}$$

$$\boxed{\text{z140}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(s)}(T_g)}{3}$$

$$\boxed{\text{z141}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,3}^{(s)}(T_g)}{3}$$

$$\boxed{\text{z142}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z143}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z144}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z145}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, c) = \frac{\sqrt{10} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(T_g) - 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) + 3\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g) \right)}{30}$$

$$\boxed{\text{z146}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, c) = \frac{\sqrt{10} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + 3\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(T_g) \right)}{30}$$

$$\boxed{\text{z147}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, c) = \frac{\sqrt{10} \left(-2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(s)}(T_g) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(T_g) \right)}{30}$$

$$\boxed{\text{z148}} \quad \mathbb{Q}_{4,1}^{(c)}(T_g, 1) = -\frac{3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(T_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(T_g)}{6}$$

$$\boxed{\text{z149}} \quad \mathbb{Q}_{4,2}^{(c)}(T_g, 1) = \frac{3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(T_g)}{6}$$

$$\boxed{\text{z150}} \quad \mathbb{Q}_{4,3}^{(c)}(T_g, 1) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(s)}(T_g)}{3}$$

$$\boxed{\text{z151}} \quad \mathbb{Q}_{4,1}^{(c)}(T_g, 2) = \frac{\sqrt{5} \left(-3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(T_g) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g) \right)}{30}$$

$$\boxed{\text{z152}} \quad \mathbb{Q}_{4,2}^{(c)}(T_g, 2) = \frac{\sqrt{5} \left(-3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(T_g) \right)}{30}$$

$$\boxed{\text{z153}} \quad \mathbb{Q}_{4,3}^{(c)}(T_g, 2) = \frac{\sqrt{5} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(s)}(T_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(T_g) \right)}{15}$$

$$\boxed{\text{z154}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z155}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\begin{aligned}
\text{z156} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3} \\
\text{z157} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, b) &= \frac{\sqrt{10}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(T_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(T_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g)\right)}{30} \\
\text{z158} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, b) &= \frac{\sqrt{10}\left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(T_g)\right)}{30} \\
\text{z159} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, b) &= \frac{\sqrt{10}\left(-2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(s)}(T_g) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(T_g)\right)}{30} \\
\text{z160} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 1) &= -\frac{3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(T_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(T_g)}{6} \\
\text{z161} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 1) &= \frac{3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(T_g)}{6} \\
\text{z162} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 1) &= \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(s)}(T_g)}{3} \\
\text{z163} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 2) &= \frac{\sqrt{5}\left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(T_g) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g)\right)}{30} \\
\text{z164} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 2) &= \frac{\sqrt{5}\left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(T_g)\right)}{30} \\
\text{z165} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 2) &= \frac{\sqrt{5}\left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(s)}(T_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(T_g)\right)}{15} \\
\text{z166} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g) &= \frac{\sqrt{6}\left(-\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g)\right)}{6} \\
\text{z167} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g) &= \frac{\sqrt{6}\left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) - \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(T_g)\right)}{6} \\
\text{z168} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g) &= \frac{\sqrt{6}\left(-\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(T_g)\right)}{6}
\end{aligned}$$

$$\boxed{\text{z169}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(T_g) = \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,1}^{(s)}(T_g)}{3}$$

$$\boxed{\text{z170}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(T_g) = \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,2}^{(s)}(T_g)}{3}$$

$$\boxed{\text{z171}} \quad \mathbb{Q}_{2,3}^{(1,1;c)}(T_g) = \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,3}^{(s)}(T_g)}{3}$$

$$\boxed{\text{z172}} \quad \mathbb{G}_{1,1}^{(c)}(T_g) = \frac{\sqrt{6}\left(-\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) + \mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g)\right)}{6}$$

$$\boxed{\text{z173}} \quad \mathbb{G}_{1,2}^{(c)}(T_g) = \frac{\sqrt{6}\left(\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) - \mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(T_g)\right)}{6}$$

$$\boxed{\text{z174}} \quad \mathbb{G}_{1,3}^{(c)}(T_g) = \frac{\sqrt{6}\left(-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(T_g)\right)}{6}$$

$$\boxed{\text{z175}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g) = \frac{\sqrt{6}\left(-\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g)\right)}{6}$$

$$\boxed{\text{z176}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g) = \frac{\sqrt{6}\left(\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) - \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(T_g)\right)}{6}$$

$$\boxed{\text{z177}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g) = \frac{\sqrt{6}\left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(T_g)\right)}{6}$$

$$\boxed{\text{z178}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z179}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z180}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z181}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, b) = \frac{\sqrt{6}\left(\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(s)}(T_g) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(T_g)\right)}{6}$$

$$\boxed{\text{z182}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, b) = \frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,3}^{(s)}(T_g) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(s)}(T_g) \right)}{6}$$

$$\boxed{\text{z183}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, b) = \frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,2}^{(s)}(T_g) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(s)}(T_g) \right)}{6}$$

$$\boxed{\text{z337}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{1,1}^{(s)}(T_u)}{3}$$

$$\boxed{\text{z338}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{1,2}^{(s)}(T_u)}{3}$$

$$\boxed{\text{z339}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{1,3}^{(s)}(T_u)}{3}$$

$$\boxed{\text{z340}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, b) = \frac{\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_3^{(s)}(A_u)}{3}$$

$$\boxed{\text{z341}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, b) = \frac{\sqrt{3} \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_3^{(s)}(A_u)}{3}$$

$$\boxed{\text{z342}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, b) = \frac{\sqrt{3} \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_3^{(s)}(A_u)}{3}$$

$$\boxed{\text{z343}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, c) = \frac{\sqrt{10} \left(-\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{1,1}^{(s)}(T_u) + 3 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{1,1}^{(s)}(T_u) + 3 \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{1,3}^{(s)}(T_u) + 3 \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{1,2}^{(s)}(T_u) \right)}{30}$$

$$\boxed{\text{z344}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, c) = \frac{\sqrt{10} \left(-\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{1,2}^{(s)}(T_u) + 3 \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{1,3}^{(s)}(T_u) - 3 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{1,2}^{(s)}(T_u) + 3 \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{30}$$

$$\boxed{\text{z345}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, c) = \frac{\sqrt{10} \left(2\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{1,3}^{(s)}(T_u) + 3 \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{1,2}^{(s)}(T_u) + 3 \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{30}$$

$$\boxed{\text{z346}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 1) = \frac{\sqrt{5} \left(-3 \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{1,1}^{(s)}(T_u) + 3\sqrt{3} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{1,1}^{(s)}(T_u) - 2\sqrt{3} \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{1,3}^{(s)}(T_u) - 2\sqrt{3} \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{1,2}^{(s)}(T_u) \right)}{30}$$

$$\boxed{\text{z347}} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 1) = -\frac{\sqrt{5} \left(3 \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{1,2}^{(s)}(T_u) + 2\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{1,3}^{(s)}(T_u) + 3\sqrt{3} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{1,2}^{(s)}(T_u) + 2\sqrt{3} \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{30}$$

$$\begin{aligned}
\text{z348} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 1) &= \frac{\sqrt{5} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,3}^{(s)}(T_u) - \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{15} \\
\text{z349} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 2) &= \frac{-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(s)}(T_u) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(s)}(T_u) - 2\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(s)}(T_u) + 2\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u)}{6} \\
\text{z350} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 2) &= \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(s)}(T_u) + 2\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(s)}(T_u) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(s)}(T_u) - 2\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u)}{6} \\
\text{z351} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 2) &= \frac{-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,3}^{(s)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u)}{3} \\
\text{z352} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_3^{(s)}(A_u)}{3} \\
\text{z353} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_3^{(s)}(A_u)}{3} \\
\text{z354} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_3^{(s)}(A_u)}{3} \\
\text{z355} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, b) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(s)}(T_u) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(s)}(T_u) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(s)}(T_u) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u) \right)}{30} \\
\text{z356} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, b) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(s)}(T_u) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(s)}(T_u) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(s)}(T_u) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{30} \\
\text{z357} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, b) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,3}^{(s)}(T_u) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{30} \\
\text{z358} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 1) &= \frac{\sqrt{5} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(s)}(T_u) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(s)}(T_u) - 2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(s)}(T_u) - 2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u) \right)}{30} \\
\text{z359} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 1) &= -\frac{\sqrt{5} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(s)}(T_u) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(s)}(T_u) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(s)}(T_u) + 2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{30} \\
\text{z360} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 1) &= \frac{\sqrt{5} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,3}^{(s)}(T_u) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{15}
\end{aligned}$$

$$\boxed{\text{z361}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2) = \frac{-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(s)}(T_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(s)}(T_u) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(s)}(T_u) + 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u)}{6}$$

$$\boxed{\text{z362}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(s)}(T_u) + 2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(s)}(T_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(s)}(T_u) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u)}{6}$$

$$\boxed{\text{z363}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2) = \frac{-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,3}^{(s)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u)}{3}$$

$$\boxed{\text{z364}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,3}^{(s)}(T_u) - \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u) \right)}{6}$$

$$\boxed{\text{z365}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u) = \frac{\sqrt{6} \left(-\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,3}^{(s)}(T_u) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{6}$$

$$\boxed{\text{z366}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{6}$$

$$\boxed{\text{z367}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2) = -\frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_3^{(s)}(A_u)}{3}$$

$$\boxed{\text{z368}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2) = -\frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_3^{(s)}(A_u)}{3}$$

$$\boxed{\text{z369}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2) = -\frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_3^{(s)}(A_u)}{3}$$

$$\boxed{\text{z370}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_u) = \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{1,1}^{(s)}(T_u)}{3}$$

$$\boxed{\text{z371}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_u) = \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{1,2}^{(s)}(T_u)}{3}$$

$$\boxed{\text{z372}} \quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_u) = \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{1,3}^{(s)}(T_u)}{3}$$

$$\boxed{\text{z373}} \quad \mathbb{G}_{2,1}^{(c)}(T_u) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(s)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(s)}(T_u) - \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(s)}(T_u) + \mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(s)}(T_u) \right)}{6}$$

$$\boxed{\text{z374}} \quad \mathbb{G}_{2,2}^{(c)}(T_u) = \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{1,2}^{(s)}(T_u) + \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{1,3}^{(s)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{1,2}^{(s)}(T_u) - \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{6}$$

$$\boxed{\text{z375}} \quad \mathbb{G}_{2,3}^{(c)}(T_u) = \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{1,2}^{(s)}(T_u) - 2 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{1,3}^{(s)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{6}$$

$$\boxed{\text{z376}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{2} \left(\sqrt{3} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,1}^{(s)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,1}^{(s)}(T_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,3}^{(s)}(T_u) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,2}^{(s)}(T_u) \right)}{6}$$

$$\boxed{\text{z377}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,2}^{(s)}(T_u) + \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,3}^{(s)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,2}^{(s)}(T_u) - \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{6}$$

$$\boxed{\text{z378}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,2}^{(s)}(T_u) - 2 \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,3}^{(s)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{6}$$

$$\boxed{\text{z379}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{1,3}^{(s)}(T_u) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{1,2}^{(s)}(T_u) \right)}{6}$$

$$\boxed{\text{z380}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{1,3}^{(s)}(T_u) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{6}$$

$$\boxed{\text{z381}} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{1,2}^{(s)}(T_u) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{1,1}^{(s)}(T_u) \right)}{6}$$

• **A;A_001_1** : 'A'-'A' bond-cluster

* bra: $\langle \frac{1}{2}, \frac{1}{2}; p |, \langle \frac{1}{2}, -\frac{1}{2}; p |, \langle \frac{3}{2}, \frac{3}{2}; p |, \langle \frac{3}{2}, \frac{1}{2}; p |, \langle \frac{3}{2}, -\frac{1}{2}; p |, \langle \frac{3}{2}, -\frac{3}{2}; p |$

* ket: $|\frac{1}{2}, \frac{1}{2}; p\rangle, |\frac{1}{2}, -\frac{1}{2}; p\rangle, |\frac{3}{2}, \frac{3}{2}; p\rangle, |\frac{3}{2}, \frac{1}{2}; p\rangle, |\frac{3}{2}, -\frac{1}{2}; p\rangle, |\frac{3}{2}, -\frac{3}{2}; p\rangle$

* wyckoff: **12b@12c**

$$\boxed{\text{z6}} \quad \mathbb{Q}_0^{(c)}(A_g, a) = \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z7}} \quad \mathbb{Q}_0^{(c)}(A_g, b) = \frac{\sqrt{5} \left(\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{5}$$

$$\boxed{\text{z8}} \quad \mathbb{Q}_0^{(c)}(A_g, c) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3}$$

$$\boxed{\text{z9}} \quad \mathbb{Q}_4^{(c)}(A_g) = \frac{\sqrt{30} \left(3 \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 2 \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - 2 \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - 2 \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{30}$$

$$\begin{aligned}
\text{z10} \quad \mathbb{Q}_0^{(1,-1;c)}(A_g, a) &= \frac{\sqrt{5} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{5} \\
\text{z11} \quad \mathbb{Q}_0^{(1,-1;c)}(A_g, b) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3} \\
\text{z12} \quad \mathbb{Q}_4^{(1,-1;c)}(A_g, a) &= \frac{\sqrt{30} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{30} \\
\text{z13} \quad \mathbb{Q}_4^{(1,-1;c)}(A_g, b) &= \frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3} \\
\text{z14} \quad \mathbb{Q}_4^{(1,-1;c)}(A_g, c) &= -\frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,1}^{(b)}(T_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,3}^{(b)}(T_g) \right)}{3} \\
\text{z15} \quad \mathbb{Q}_0^{(1,0;c)}(A_g) &= \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{2,3}^{(b)}(T_g) \right)}{3} \\
\text{z16} \quad \mathbb{Q}_0^{(1,1;c)}(A_g, a) &= \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_0^{(b)}(A_g) \\
\text{z17} \quad \mathbb{Q}_0^{(1,1;c)}(A_g, b) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3} \\
\text{z18} \quad \mathbb{G}_3^{(c)}(A_g, a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2} \\
\text{z19} \quad \mathbb{G}_3^{(c)}(A_g, b) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{2,3}^{(b)}(T_g) \right)}{3} \\
\text{z20} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2} \\
\text{z21} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, b) &= -\frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3} \\
\text{z22} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, c) &= -\frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,1}^{(b)}(T_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,3}^{(b)}(T_g) \right)}{3}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z23}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, d) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{2,3}^{(b)}(T_g) \right)}{3} \\
\boxed{\text{z24}} \quad \mathbb{G}_3^{(1,0;c)}(A_g, a) &= \frac{\sqrt{3} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{3} \\
\boxed{\text{z25}} \quad \mathbb{G}_3^{(1,0;c)}(A_g, b) &= \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3} \\
\boxed{\text{z26}} \quad \mathbb{G}_3^{(1,1;c)}(A_g) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{2,3}^{(b)}(T_g) \right)}{3} \\
\boxed{\text{z32}} \quad \mathbb{Q}_3^{(c)}(A_u, a) &= \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{3} \\
\boxed{\text{z33}} \quad \mathbb{Q}_3^{(c)}(A_u, b) &= -\frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{3} \\
\boxed{\text{z34}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, a) &= \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{3} \\
\boxed{\text{z35}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, b) &= -\frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{3} \\
\boxed{\text{z36}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, c) &= -\frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3} \\
\boxed{\text{z37}} \quad \mathbb{Q}_3^{(1,0;c)}(A_u, a) &= \frac{\sqrt{2} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{M}_{2,2}^{(b)}(E_u) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{M}_{2,1}^{(b)}(E_u) \right)}{2} \\
\boxed{\text{z38}} \quad \mathbb{Q}_3^{(1,0;c)}(A_u, b) &= \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3} \\
\boxed{\text{z39}} \quad \mathbb{G}_0^{(c)}(A_u) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3} \\
\boxed{\text{z40}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u, a) &= \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_3^{(b)}(A_u)
\end{aligned}$$

$$\boxed{\text{z41}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u, b) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z42}} \quad \mathbb{G}_4^{(1,-1;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z43}} \quad \mathbb{G}_0^{(1,0;c)}(A_u, a) = \frac{\sqrt{3} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z44}} \quad \mathbb{G}_0^{(1,0;c)}(A_u, b) = \frac{\sqrt{2} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{M}_{2,1}^{(b)}(E_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{M}_{2,2}^{(b)}(E_u) \right)}{2}$$

$$\boxed{\text{z45}} \quad \mathbb{G}_4^{(1,0;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{3}$$

$$\boxed{\text{z46}} \quad \mathbb{G}_0^{(1,1;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z57}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z58}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z59}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, b) = \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z60}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, b) = \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z61}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, c) = \frac{\sqrt{7} \left(2 \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 2 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - 2 \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{14}$$

$$\boxed{\text{z62}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, c) = \frac{\sqrt{7} \left(-2 \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3} \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 2 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3} \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{14}$$

$$\boxed{\text{z63}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, d) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + 2 \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{6}$$

$$\boxed{z64} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, d) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{2}$$

$$\boxed{z65} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, e) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{2,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{2,2}^{(b)}(T_g)}{2}$$

$$\boxed{z66} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, e) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{2,1}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{2,2}^{(b)}(T_g) - 2\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{2,3}^{(b)}(T_g) \right)}{6}$$

$$\boxed{z67} \quad \mathbb{Q}_{4,1}^{(c)}(E_g) = \frac{\sqrt{21} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - 2\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 4\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{42}$$

$$\boxed{z68} \quad \mathbb{Q}_{4,2}^{(c)}(E_g) = \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{14}$$

$$\boxed{z69} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{z70} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{z71} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) = \frac{\sqrt{7} \left(2\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{14}$$

$$\boxed{z72} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) = \frac{\sqrt{7} \left(-2\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{14}$$

$$\boxed{z73} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, c) = \frac{\sqrt{14} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,3}^{(b)}(T_g) \right)}{28}$$

$$\boxed{z74} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, c) = \frac{\sqrt{14} \left(9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,3}^{(b)}(T_g) \right)}{84}$$

$$\boxed{z75} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, d) = \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,1}^{(b)}(T_g) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,1}^{(b)}(T_g) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,2}^{(b)}(T_g) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,2}^{(b)}(T_g) + 2\sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,3}^{(b)}(T_g) \right)}{12}$$

$$\boxed{z76} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, d) = \frac{\sqrt{2} \left(-\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,1}^{(b)}(T_g) + \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,1}^{(b)}(T_g) - \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,2}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,2}^{(b)}(T_g) + 2\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,3}^{(b)}(T_g) \right)}{12}$$

$$\begin{aligned}
\boxed{\text{z77}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, e) &= \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z78}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, e) &= \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g)}{2} \\
\boxed{\text{z79}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, f) &= \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g)}{2} \\
\boxed{\text{z80}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, f) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g) - 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{2,3}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z81}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, a) &= \frac{\sqrt{21} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 4\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{42} \\
\boxed{\text{z82}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, a) &= \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3} \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{14} \\
\boxed{\text{z83}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, b) &= \frac{\sqrt{14} \left(-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g) + 2\sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{84} \\
\boxed{\text{z84}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, b) &= \frac{\sqrt{14} \left(\sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) + \sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) + \sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g) - 2\sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{28} \\
\boxed{\text{z85}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, c) &= \frac{\sqrt{2} \left(-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,1}^{(b)}(T_g) + \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,1}^{(b)}(T_g) + \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,2}^{(b)}(T_g) - 2\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,3}^{(b)}(T_g) \right)}{12} \\
\boxed{\text{z86}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, c) &= \frac{\sqrt{2} \left(-\sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,1}^{(b)}(T_g) - \sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,1}^{(b)}(T_g) - \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,2}^{(b)}(T_g) + \sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,2}^{(b)}(T_g) + 2\sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,3}^{(b)}(T_g) \right)}{12} \\
\boxed{\text{z87}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, a) &= \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g)}{2} \\
\boxed{\text{z88}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, a) &= \frac{\sqrt{3} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - 2\mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z89}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, b) &= \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g)}{2}
\end{aligned}$$

$$\boxed{\text{z90}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, b) = \frac{\sqrt{3} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z91}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, c) = \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{2,3}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z92}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, c) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g)}{2}$$

$$\boxed{\text{z93}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z94}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z95}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, b) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + 2\mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z96}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, b) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g)}{2}$$

$$\boxed{\text{z97}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, c) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g)}{2}$$

$$\boxed{\text{z98}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, c) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g) - 2\mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{2,3}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z109}} \quad \mathbb{Q}_{5,1}^{(c)}(E_u) = \frac{\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{2}$$

$$\boxed{\text{z110}} \quad \mathbb{Q}_{5,2}^{(c)}(E_u) = \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) - 2\mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z111}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u, a) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{2}$$

$$\boxed{\text{z112}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z113}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u, b) = \frac{\sqrt{2}\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z114}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u, b) = -\frac{\sqrt{2}\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{M}_{2,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z115}} \quad \mathbb{Q}_{5,1}^{(1,0;c)}(E_u) = \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_3^{(b)}(A_u)}{2}$$

$$\boxed{\text{z116}} \quad \mathbb{Q}_{5,2}^{(1,0;c)}(E_u) = -\frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_3^{(b)}(A_u)}{2}$$

$$\boxed{\text{z117}} \quad \mathbb{G}_{2,1}^{(c)}(E_u, a) = \frac{-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z118}} \quad \mathbb{G}_{2,2}^{(c)}(E_u, a) = \frac{\sqrt{3}\left(-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)\right)}{6}$$

$$\boxed{\text{z119}} \quad \mathbb{G}_{2,1}^{(c)}(E_u, b) = \frac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)\right)}{6}$$

$$\boxed{\text{z120}} \quad \mathbb{G}_{2,2}^{(c)}(E_u, b) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z121}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z122}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{3}\left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)\right)}{6}$$

$$\boxed{\text{z123}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\sqrt{14}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u)\right)}{28}$$

$$\boxed{\text{z124}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = \frac{\sqrt{14}\left(9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u)\right)}{84}$$

$$\boxed{\text{z125}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, c) = \frac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)\right)}{6}$$

$$\boxed{\text{z126}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, c) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z127}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{14} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{84}$$

$$\boxed{\text{z128}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{14} \left(\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) - 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{28}$$

$$\boxed{\text{z129}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u, a) = \frac{\sqrt{3} \left(-\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z130}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u, a) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z131}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u, b) = \frac{\sqrt{3} \left(-\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 2\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z132}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u, b) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{2}$$

$$\boxed{\text{z133}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u, c) = \frac{\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{2,1}^{(b)}(E_u) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z134}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u, c) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{2,2}^{(b)}(E_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{2,1}^{(b)}(E_u)}{2}$$

$$\boxed{\text{z135}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u, d) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z136}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u, d) = \frac{\sqrt{3} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z137}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z138}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z184}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z185}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z186}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z187}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z188}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z189}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z190}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, c) = \frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 3\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{42}$$

$$\boxed{\text{z191}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, c) = \frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{42}$$

$$\boxed{\text{z192}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, c) = \frac{\sqrt{14} \left(-2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 2\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42}$$

$$\boxed{\text{z193}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, d) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z194}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, d) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z195}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, d) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z196}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, e) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{2,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{2,2}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z197}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, e) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{2,3}^{(b)}(T_g) - \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z198}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, e) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z199}} \quad \mathbb{Q}_{4,1}^{(c)}(T_g, 1) = -\frac{\sqrt{2} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{12}$$

$$\boxed{\text{z200}} \quad \mathbb{Q}_{4,2}^{(c)}(T_g, 1) = \frac{\sqrt{2} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12}$$

$$\boxed{\text{z201}} \quad \mathbb{Q}_{4,3}^{(c)}(T_g, 1) = \frac{\sqrt{6} \left(\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6}$$

$$\boxed{\text{z202}} \quad \mathbb{Q}_{4,1}^{(c)}(T_g, 2) = \frac{\sqrt{14} \left(-3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 3\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{84}$$

$$\boxed{\text{z203}} \quad \mathbb{Q}_{4,2}^{(c)}(T_g, 2) = \frac{\sqrt{14} \left(-3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - 3\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{84}$$

$$\boxed{\text{z204}} \quad \mathbb{Q}_{4,3}^{(c)}(T_g, 2) = \frac{\sqrt{14} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42}$$

$$\boxed{\text{z205}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z206}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z207}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z208}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{42}$$

$$\begin{aligned}
\text{z209} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, b) &= \frac{\sqrt{14} \left(\sqrt{3} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3 \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 3 \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + \sqrt{3} \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3 \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + 3 \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{42} \\
\text{z210} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, b) &= \frac{\sqrt{14} \left(-2\sqrt{3} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 3 \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3 \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 2\sqrt{3} \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42} \\
\text{z211} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, c) &= \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) + \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{21} \\
\text{z212} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, c) &= \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) + \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{M}_{1,2}^{(b)}(T_g) \right)}{21} \\
\text{z213} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, c) &= \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) + \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{21} \\
\text{z214} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, d) &= \frac{\sqrt{6} \left(-\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,3}^{(b)}(T_g) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,2}^{(b)}(T_g) \right)}{6} \\
\text{z215} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, d) &= \frac{\sqrt{6} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,3}^{(b)}(T_g) - \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,1}^{(b)}(T_g) \right)}{6} \\
\text{z216} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, d) &= \frac{\sqrt{6} \left(-\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,1}^{(b)}(T_g) \right)}{6} \\
\text{z217} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, e) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6} \\
\text{z218} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, e) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6} \\
\text{z219} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, e) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6} \\
\text{z220} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, f) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{2,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g) \right)}{6}
\end{aligned}$$

$$\begin{aligned}
\text{z221} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, f) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{2,3}^{(b)}(T_g) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) \right)}{6} \\
\text{z222} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, f) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) \right)}{6} \\
\text{z223} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 1a) &= -\frac{\sqrt{2} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{12} \\
\text{z224} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 1a) &= \frac{\sqrt{2} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12} \\
\text{z225} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 1a) &= \frac{\sqrt{6} \left(\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6} \\
\text{z226} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 1b) &= \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g)}{12} \\
\text{z227} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 1b) &= \frac{-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g)}{12} \\
\text{z228} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 1b) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g)}{12} \\
\text{z229} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 1c) &= \frac{-3\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,3}^{(b)}(T_g) + 3\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,3}^{(b)}(T_g) - 3\sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,2}^{(b)}(T_g) - 3\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,2}^{(b)}(T_g) - 4\sqrt{3}\mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{2,1}^{(b)}(T_g)}{24} \\
\text{z230} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 1c) &= \frac{-3\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,3}^{(b)}(T_g) - 3\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,3}^{(b)}(T_g) - 3\sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,1}^{(b)}(T_g) + 3\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,1}^{(b)}(T_g) - 4\sqrt{3}\mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{2,2}^{(b)}(T_g)}{24} \\
\text{z231} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 1c) &= \frac{-3\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,2}^{(b)}(T_g) + 3\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,2}^{(b)}(T_g) - 3\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,1}^{(b)}(T_g) - 3\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,1}^{(b)}(T_g) - 4\sqrt{3}\mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{2,3}^{(b)}(T_g)}{24} \\
\text{z232} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 2a) &= \frac{\sqrt{14} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{84}
\end{aligned}$$

$$\begin{aligned}
\boxed{z233} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 2a) &= \frac{\sqrt{14} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{84} \\
\boxed{z234} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 2a) &= \frac{\sqrt{14} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42} \\
\boxed{z235} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 2b) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,3}^{(b)}(T_g) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g) + 9\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g) + 12\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{84} \\
\boxed{z236} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 2b) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,3}^{(b)}(T_g) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g) - 9\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g) + 12\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{84} \\
\boxed{z237} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 2b) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g) - 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g) + 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g) + 12\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{M}_{1,3}^{(b)}(T_g) \right)}{84} \\
\boxed{z238} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 2c) &= -\frac{\sqrt{6} \left(\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,3}^{(b)}(T_g) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,2}^{(b)}(T_g) \right)}{6} \\
\boxed{z239} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 2c) &= -\frac{\sqrt{6} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,3}^{(b)}(T_g) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,1}^{(b)}(T_g) \right)}{6} \\
\boxed{z240} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 2c) &= -\frac{\sqrt{6} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,1}^{(b)}(T_g) \right)}{6} \\
\boxed{z241} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{6} \\
\boxed{z242} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{2} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{6} \\
\boxed{z243} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{2} \left(-\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + 2\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6} \\
\boxed{z244} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g, b) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6}
\end{aligned}$$

$$\begin{aligned}
\text{z245} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g, b) &= \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) - \mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6} \\
\text{z246} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g, b) &= \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) - 2\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6} \\
\text{z247} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g, c) &= \frac{\sqrt{10} \left(\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{2,1}^{(b)}(T_g) - 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{2,1}^{(b)}(T_g) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{2,3}^{(b)}(T_g) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(T_g) \right)}{30} \\
\text{z248} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g, c) &= \frac{\sqrt{10} \left(\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{2,2}^{(b)}(T_g) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{2,3}^{(b)}(T_g) + 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{2,2}^{(b)}(T_g) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(T_g) \right)}{30} \\
\text{z249} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g, c) &= \frac{\sqrt{10} \left(-2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{2,3}^{(b)}(T_g) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(T_g) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(T_g) \right)}{30} \\
\text{z250} \quad \mathbb{Q}_{4,1}^{(1,0;c)}(T_g, 1) &= -\frac{3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{2,1}^{(b)}(T_g) + \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{2,1}^{(b)}(T_g)}{6} \\
\text{z251} \quad \mathbb{Q}_{4,2}^{(1,0;c)}(T_g, 1) &= \frac{3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{2,2}^{(b)}(T_g) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{2,2}^{(b)}(T_g)}{6} \\
\text{z252} \quad \mathbb{Q}_{4,3}^{(1,0;c)}(T_g, 1) &= \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{2,3}^{(b)}(T_g)}{3} \\
\text{z253} \quad \mathbb{Q}_{4,1}^{(1,0;c)}(T_g, 2) &= \frac{\sqrt{5} \left(-3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{2,1}^{(b)}(T_g) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{2,1}^{(b)}(T_g) + 2\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{2,3}^{(b)}(T_g) + 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(T_g) \right)}{30} \\
\text{z254} \quad \mathbb{Q}_{4,2}^{(1,0;c)}(T_g, 2) &= \frac{\sqrt{5} \left(-3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{2,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{2,3}^{(b)}(T_g) - 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{2,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(T_g) \right)}{30} \\
\text{z255} \quad \mathbb{Q}_{4,3}^{(1,0;c)}(T_g, 2) &= \frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(T_g) + \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(T_g) \right)}{15} \\
\text{z256} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3} \\
\text{z257} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z258}} \quad \mathbb{Q}_{2,3}^{(1,1;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3} \\
\boxed{\text{z259}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(T_g, b) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z260}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(T_g, b) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z261}} \quad \mathbb{Q}_{2,3}^{(1,1;c)}(T_g, b) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z262}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(T_g, c) &= \frac{\sqrt{6}\left(-\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{2,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z263}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(T_g, c) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{2,3}^{(b)}(T_g) - \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z264}} \quad \mathbb{Q}_{2,3}^{(1,1;c)}(T_g, c) &= \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z265}} \quad \mathbb{G}_{1,1}^{(c)}(T_g, a) &= \frac{\sqrt{10}\left(-3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)\right)}{30} \\
\boxed{\text{z266}} \quad \mathbb{G}_{1,2}^{(c)}(T_g, a) &= \frac{\sqrt{10}\left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)\right)}{30} \\
\boxed{\text{z267}} \quad \mathbb{G}_{1,3}^{(c)}(T_g, a) &= \frac{\sqrt{30}\left(-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 2\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 2\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)\right)}{30} \\
\boxed{\text{z268}} \quad \mathbb{G}_{1,1}^{(c)}(T_g, b) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) - \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z269}} \quad \mathbb{G}_{1,2}^{(c)}(T_g, b) &= \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z270}} \quad \mathbb{G}_{1,3}^{(c)}(T_g, b) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)\right)}{6}
\end{aligned}$$

$$\boxed{\text{z271}} \quad \mathbb{G}_{1,1}^{(c)}(T_g, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{2,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z272}} \quad \mathbb{G}_{1,2}^{(c)}(T_g, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{2,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z273}} \quad \mathbb{G}_{1,3}^{(c)}(T_g, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z274}} \quad \mathbb{G}_{3,1}^{(c)}(T_g, 1) = \frac{\sqrt{10} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 3\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3} \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 4\sqrt{3} \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 4\sqrt{3} \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{60}$$

$$\boxed{\text{z275}} \quad \mathbb{G}_{3,2}^{(c)}(T_g, 1) = \frac{\sqrt{10} \left(-3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 4\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3} \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - 4\sqrt{3} \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{60}$$

$$\boxed{\text{z276}} \quad \mathbb{G}_{3,3}^{(c)}(T_g, 1) = \frac{\sqrt{30} \left(-2\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 2\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{30}$$

$$\boxed{\text{z277}} \quad \mathbb{G}_{3,1}^{(c)}(T_g, 2) = \frac{\sqrt{2} \left(\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3} \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{12}$$

$$\boxed{\text{z278}} \quad \mathbb{G}_{3,2}^{(c)}(T_g, 2) = \frac{\sqrt{2} \left(\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - \sqrt{3} \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12}$$

$$\boxed{\text{z279}} \quad \mathbb{G}_{3,3}^{(c)}(T_g, 2) = \frac{\sqrt{6} \left(-\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{6}$$

$$\boxed{\text{z280}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{10} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3} \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3} \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3} \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3} \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{30}$$

$$\boxed{\text{z281}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{10} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + \sqrt{3} \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) - \sqrt{3} \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3} \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3} \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{30}$$

$$\boxed{\text{z282}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{30} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 2\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{30}$$

$$\boxed{\text{z283}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{79} \left(-63\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,3}^{(b)}(T_g) - 13\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,3}^{(b)}(T_g) - 63\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,2}^{(b)}(T_g) + 13\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,2}^{(b)}(T_g) + 12\sqrt{15}\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{2,1}^{(b)}(T_g) \right)}{1896}$$

$$\boxed{\text{z284}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{79} \left(-63\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,3}^{(b)}(T_g) + 13\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,3}^{(b)}(T_g) - 63\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,1}^{(b)}(T_g) - 13\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,1}^{(b)}(T_g) + 12\sqrt{15}\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{2,2}^{(b)}(T_g) \right)}{1896}$$

$$\boxed{\text{z285}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{79} \left(-63\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,2}^{(b)}(T_g) - 13\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,2}^{(b)}(T_g) - 63\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,1}^{(b)}(T_g) + 13\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,1}^{(b)}(T_g) + 12\sqrt{15}\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{2,3}^{(b)}(T_g) \right)}{1896}$$

$$\boxed{\text{z286}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z287}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, c) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z288}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z289}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, d) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z290}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, d) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z291}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, d) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z292}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 1a) = \frac{\sqrt{10} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 4\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{60}$$

$$\boxed{\text{z293}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 1a) = \frac{\sqrt{10} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{60}$$

$$\begin{aligned}
\text{z294} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 1a) &= \frac{\sqrt{30} \left(-2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{30} \\
\text{z295} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 1b) &= \frac{-3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,3}^{(b)}(T_g) + 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g)}{12} \\
\text{z296} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 1b) &= \frac{3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,3}^{(b)}(T_g) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g)}{12} \\
\text{z297} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 1b) &= \frac{-3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g)}{12} \\
\text{z298} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 1c) &= \frac{\sqrt{79} \left(\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,3}^{(b)}(T_g) - 27\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,3}^{(b)}(T_g) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,2}^{(b)}(T_g) + 27\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,2}^{(b)}(T_g) - 48\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{2,1}^{(b)}(T_g) \right)}{948} \\
\text{z299} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 1c) &= \frac{\sqrt{79} \left(\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,3}^{(b)}(T_g) + 27\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,3}^{(b)}(T_g) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,1}^{(b)}(T_g) - 27\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,1}^{(b)}(T_g) - 48\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{2,2}^{(b)}(T_g) \right)}{948} \\
\text{z300} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 1c) &= \frac{\sqrt{79} \left(\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,2}^{(b)}(T_g) - 27\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,2}^{(b)}(T_g) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,1}^{(b)}(T_g) + 27\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,1}^{(b)}(T_g) - 48\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{2,3}^{(b)}(T_g) \right)}{948} \\
\text{z301} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 2a) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{12} \\
\text{z302} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 2a) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12} \\
\text{z303} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 2a) &= \frac{\sqrt{6} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{6} \\
\text{z304} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 2b) &= \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,3}^{(b)}(T_g) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g) - \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{12} \\
\text{z305} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 2b) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,3}^{(b)}(T_g) - \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,3}^{(b)}(T_g) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{12} \\
\text{z306} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 2b) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{12}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z307}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z308}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z309}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z310}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, b) &= \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 3\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)\right)}{30} \\
\boxed{\text{z311}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, b) &= \frac{\sqrt{10}\left(3\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) - \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)\right)}{30} \\
\boxed{\text{z312}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, b) &= \frac{\sqrt{10}\left(3\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + 2\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)\right)}{30} \\
\boxed{\text{z313}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, c) &= \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) + 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)\right)}{30} \\
\boxed{\text{z314}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, c) &= \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) - 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)\right)}{30} \\
\boxed{\text{z315}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, c) &= \frac{\sqrt{10}\left(2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,3}^{(b)}(T_g) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)\right)}{30} \\
\boxed{\text{z316}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, d) &= \frac{\sqrt{6}\left(-\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{2,3}^{(b)}(T_g) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z317}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, d) &= \frac{\sqrt{6}\left(\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{2,3}^{(b)}(T_g) - \mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z318}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, d) &= \frac{\sqrt{6}\left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z319}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(T_g, 1a) &= \frac{\sqrt{5}\left(-3\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - 2\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) - 2\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)\right)}{30}
\end{aligned}$$

$$\begin{aligned}
\text{z320} \quad \mathbb{G}_{3,2}^{(1,0;c)}(T_g, 1a) &= -\frac{\sqrt{5} \left(2\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{30} \\
\text{z321} \quad \mathbb{G}_{3,3}^{(1,0;c)}(T_g, 1a) &= \frac{\sqrt{5} \left(-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{15} \\
\text{z322} \quad \mathbb{G}_{3,1}^{(1,0;c)}(T_g, 1b) &= \frac{\sqrt{5} \left(-3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) - 2\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) - 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{30} \\
\text{z323} \quad \mathbb{G}_{3,2}^{(1,0;c)}(T_g, 1b) &= -\frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{30} \\
\text{z324} \quad \mathbb{G}_{3,3}^{(1,0;c)}(T_g, 1b) &= \frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{15} \\
\text{z325} \quad \mathbb{G}_{3,1}^{(1,0;c)}(T_g, 2a) &= \frac{-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) - 2\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{6} \\
\text{z326} \quad \mathbb{G}_{3,2}^{(1,0;c)}(T_g, 2a) &= \frac{-2\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{6} \\
\text{z327} \quad \mathbb{G}_{3,3}^{(1,0;c)}(T_g, 2a) &= \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3} \\
\text{z328} \quad \mathbb{G}_{3,1}^{(1,0;c)}(T_g, 2b) &= \frac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) - 2\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} \\
\text{z329} \quad \mathbb{G}_{3,2}^{(1,0;c)}(T_g, 2b) &= \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} \\
\text{z330} \quad \mathbb{G}_{3,3}^{(1,0;c)}(T_g, 2b) &= \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{3} \\
\text{z331} \quad \mathbb{G}_{1,1}^{(1,1;c)}(T_g, a) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) - \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6} \\
\text{z332} \quad \mathbb{G}_{1,2}^{(1,1;c)}(T_g, a) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z333}} \quad \mathbb{G}_{1,3}^{(1,1;c)}(T_g, a) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z334}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(T_g, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{2,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z335}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(T_g, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{2,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z336}} \quad \mathbb{G}_{1,3}^{(1,1;c)}(T_g, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z382}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{1,1}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z383}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{1,2}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z384}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z385}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, b) = \frac{\sqrt{10} \left(-\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{1,1}^{(b)}(T_u) + 3 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{1,1}^{(b)}(T_u) + 3 \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) + 3 \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z386}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, b) = \frac{\sqrt{10} \left(-\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + 3 \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) - 3 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + 3 \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z387}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, b) = \frac{\sqrt{10} \left(2\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{1,3}^{(b)}(T_u) + 3 \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + 3 \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z388}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, c) = \frac{\sqrt{5} \left(-3 \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 3\sqrt{3} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) - 2\sqrt{3} \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) - 2\sqrt{3} \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{30}$$

$$\boxed{\text{z389}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, c) = -\frac{\sqrt{5} \left(3 \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 2\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 3\sqrt{3} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 2\sqrt{3} \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{30}$$

$$\boxed{\text{z390}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, c) = \frac{\sqrt{5} \left(3 \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) - \sqrt{3} \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) - \sqrt{3} \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{15}$$

$$\boxed{\text{z391}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, d) = \frac{-\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) + 3\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6}$$

$$\boxed{\text{z392}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, d) = -\frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) + 3\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6}$$

$$\boxed{\text{z393}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, d) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u)}{3}$$

$$\boxed{\text{z394}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, e) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z395}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, e) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z396}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, e) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z397}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 1a) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{3}$$

$$\boxed{\text{z398}} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 1a) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{3}$$

$$\boxed{\text{z399}} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 1a) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1)}{3}$$

$$\boxed{\text{z400}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 1b) = \frac{\sqrt{5} \left(-3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - 2\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) - 2\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z401}} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 1b) = -\frac{\sqrt{5} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z402}} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 1b) = \frac{\sqrt{5} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u) - \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{15}$$

$$\boxed{\text{z403}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 1c) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{30}$$

$$\begin{aligned}
\text{z404} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 1c) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{30} \\
\text{z405} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 1c) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{30} \\
\text{z406} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 2a) &= \frac{-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - 2\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + 2\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} \\
\text{z407} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 2a) &= \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) - 2\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} \\
\text{z408} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 2a) &= \frac{-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3} \\
\text{z409} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 2b) &= \frac{\sqrt{2} \left(6\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{18} \\
\text{z410} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 2b) &= \frac{\sqrt{2} \left(-6\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - \sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{18} \\
\text{z411} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 2b) &= \frac{\sqrt{6} \left(-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - 4\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{18} \\
\text{z412} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 2c) &= -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\text{z413} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 2c) &= -\frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\text{z414} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 2c) &= -\frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\text{z415} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 2d) &= -\frac{3\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6} \\
\text{z416} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 2d) &= \frac{3\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6}
\end{aligned}$$

$$\begin{aligned}
\text{z417} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 2d) &= \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{3} \\
\text{z418} \quad \mathbb{Q}_{5,1}^{(c)}(T_u, 3) &= \frac{3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 4\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - 4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{18} \\
\text{z419} \quad \mathbb{Q}_{5,2}^{(c)}(T_u, 3) &= \frac{-3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - 4\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{18} \\
\text{z420} \quad \mathbb{Q}_{5,3}^{(c)}(T_u, 3) &= \frac{\sqrt{3}\left(2\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - 2\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)\right)}{9} \\
\text{z421} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)\right)}{30} \\
\text{z422} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)\right)}{30} \\
\text{z423} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{10}\left(2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)\right)}{30} \\
\text{z424} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, b) &= \frac{\sqrt{5}\left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) - 2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - 2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)\right)}{30} \\
\text{z425} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, b) &= -\frac{\sqrt{5}\left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)\right)}{30} \\
\text{z426} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, b) &= \frac{\sqrt{5}\left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)\right)}{15} \\
\text{z427} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, c) &= \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\text{z428} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, c) &= \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\text{z429} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, c) &= \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_3^{(b)}(A_u)}{3}
\end{aligned}$$

$$\begin{aligned}
\text{z430} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, d) &= \frac{\sqrt{14} \left(-3\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,1}^{(b)}(E_u) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,2}^{(b)}(E_u) - 3\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,1}^{(b)}(E_u) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,2}^{(b)}(E_u) \right)}{84} \\
\text{z431} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, d) &= \frac{\sqrt{14} \left(-3\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,1}^{(b)}(E_u) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,2}^{(b)}(E_u) + 3\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,1}^{(b)}(E_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,2}^{(b)}(E_u) \right)}{84} \\
\text{z432} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, d) &= \frac{\sqrt{42} \left(3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,1}^{(b)}(E_u) + \sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,2}^{(b)}(E_u) \right)}{42} \\
\text{z433} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, e) &= \frac{-\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) + 3\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6} \\
\text{z434} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, e) &= -\frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) + 3\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6} \\
\text{z435} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, e) &= \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u)}{3} \\
\text{z436} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, f) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6} \\
\text{z437} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, f) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\text{z438} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, f) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\text{z439} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 1a) &= \frac{\sqrt{5} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - 2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) - 2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{30} \\
\text{z440} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 1a) &= -\frac{\sqrt{5} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{30} \\
\text{z441} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 1a) &= \frac{\sqrt{5} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{15} \\
\text{z442} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 1b) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{30}
\end{aligned}$$

$$\begin{aligned}
\text{z443} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 1b) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{30} \\
\text{z444} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 1b) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{30} \\
\text{z445} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 1c) &= \frac{\sqrt{14} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,1}^{(b)}(E_u) + 3\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,2}^{(b)}(E_u) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,2}^{(b)}(E_u) \right)}{84} \\
\text{z446} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 1c) &= \frac{\sqrt{14} \left(-\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,1}^{(b)}(E_u) - 3\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,2}^{(b)}(E_u) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,1}^{(b)}(E_u) + 3\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,2}^{(b)}(E_u) \right)}{84} \\
\text{z447} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 1c) &= \frac{\sqrt{42} \left(\sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,1}^{(b)}(E_u) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,2}^{(b)}(E_u) \right)}{42} \\
\text{z448} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 1d) &= \frac{-3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} \\
\text{z449} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 1d) &= \frac{3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} \\
\text{z450} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 1d) &= \frac{-3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} \\
\text{z451} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2a) &= \frac{-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} \\
\text{z452} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2a) &= \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} \\
\text{z453} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2a) &= \frac{-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3} \\
\text{z454} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2b) &= \frac{\sqrt{2} \left(6\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{18} \\
\text{z455} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2b) &= \frac{\sqrt{2} \left(-6\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - \sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{18}
\end{aligned}$$

$$\begin{aligned}
\text{z456} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2b) &= \frac{\sqrt{6} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) - 4\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{18} \\
\text{z457} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2c) &= -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\text{z458} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2c) &= -\frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\text{z459} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2c) &= -\frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\text{z460} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2d) &= \frac{3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,2}^{(b)}(E_u)}{6} \\
\text{z461} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2d) &= \frac{-3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,2}^{(b)}(E_u)}{6} \\
\text{z462} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2d) &= -\frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{2,2}^{(b)}(E_u)}{3} \\
\text{z463} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2e) &= \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} \\
\text{z464} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2e) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} \\
\text{z465} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2e) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} \\
\text{z466} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2f) &= -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\text{z467} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2f) &= -\frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\text{z468} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2f) &= -\frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_3^{(b)}(A_u)}{3}
\end{aligned}$$

$$\boxed{\text{z469}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2g) = -\frac{3\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6}$$

$$\boxed{\text{z470}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2g) = \frac{3\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6}$$

$$\boxed{\text{z471}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2g) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{3}$$

$$\boxed{\text{z472}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(T_u, 3a) = \frac{3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 4\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{18}$$

$$\boxed{\text{z473}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(T_u, 3a) = \frac{-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - 4\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{18}$$

$$\boxed{\text{z474}} \quad \mathbb{Q}_{5,3}^{(1,-1;c)}(T_u, 3a) = \frac{\sqrt{3} \left(2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{9}$$

$$\boxed{\text{z475}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(T_u, 3b) = \frac{-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,1}^{(b)}(E_u) + 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,2}^{(b)}(E_u)}{6}$$

$$\boxed{\text{z476}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(T_u, 3b) = -\frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,1}^{(b)}(E_u) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,2}^{(b)}(E_u)}{6}$$

$$\boxed{\text{z477}} \quad \mathbb{Q}_{5,3}^{(1,-1;c)}(T_u, 3b) = \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{2,1}^{(b)}(E_u)}{3}$$

$$\boxed{\text{z478}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, a) = \frac{\sqrt{6} \left(\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) - \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z479}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, a) = \frac{\sqrt{6} \left(-\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z480}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, a) = \frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z481}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, b) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_3^{(b)}(A_u)}{3}$$

$$\begin{aligned}
\boxed{\text{z482}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, b) &= \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\boxed{\text{z483}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, b) &= \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\boxed{\text{z484}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, c) &= \frac{3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6} \\
\boxed{\text{z485}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, c) &= \frac{-3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) + \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6} \\
\boxed{\text{z486}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, c) &= -\frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{3} \\
\boxed{\text{z487}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, d) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) + 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z488}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, d) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z489}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, d) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z490}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 1a) &= \frac{\sqrt{6} \left(-\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{6} \\
\boxed{\text{z491}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 1a) &= \frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6} \\
\boxed{\text{z492}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 1a) &= \frac{\sqrt{6} \left(-\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6} \\
\boxed{\text{z493}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 1b) &= \frac{\sqrt{5} \left(-3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) - 2\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z494}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 1b) &= -\frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30}
\end{aligned}$$

$$\begin{aligned}
\text{z495} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 1b) &= \frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{15} \\
\text{z496} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2a) &= -\frac{\sqrt{6} \left(\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{6} \\
\text{z497} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2a) &= -\frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6} \\
\text{z498} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2a) &= -\frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6} \\
\text{z499} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2b) &= \frac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6} \\
\text{z500} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2b) &= -\frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6} \\
\text{z501} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2b) &= \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u)}{3} \\
\text{z502} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2c) &= \frac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) - 2\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} \\
\text{z503} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2c) &= \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} \\
\text{z504} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2c) &= \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} \\
\text{z505} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3} \\
\text{z506} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3} \\
\text{z507} \quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z508}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_u, b) &= \frac{-\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) + 3\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6} \\
\boxed{\text{z509}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_u, b) &= -\frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) + 3\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6} \\
\boxed{\text{z510}} \quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_u, b) &= \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u)}{3} \\
\boxed{\text{z511}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_u, c) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6} \\
\boxed{\text{z512}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_u, c) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\boxed{\text{z513}} \quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_u, c) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\boxed{\text{z514}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(T_u, 1) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{3} \\
\boxed{\text{z515}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(T_u, 1) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{3} \\
\boxed{\text{z516}} \quad \mathbb{Q}_{3,3}^{(1,1;c)}(T_u, 1) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1)}{3} \\
\boxed{\text{z517}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(T_u, 2a) &= -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\boxed{\text{z518}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(T_u, 2a) &= -\frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\boxed{\text{z519}} \quad \mathbb{Q}_{3,3}^{(1,1;c)}(T_u, 2a) &= -\frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_3^{(b)}(A_u)}{3} \\
\boxed{\text{z520}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(T_u, 2b) &= -\frac{3\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) + \sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6}
\end{aligned}$$

$$\boxed{\text{z521}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(T_u, 2b) = \frac{3\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{2,1}^{(b)}(E_u) - \sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{6}$$

$$\boxed{\text{z522}} \quad \mathbb{Q}_{3,3}^{(1,1;c)}(T_u, 2b) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{2,2}^{(b)}(E_u)}{3}$$

$$\boxed{\text{z523}} \quad \mathbb{G}_{2,1}^{(c)}(T_u, a) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z524}} \quad \mathbb{G}_{2,2}^{(c)}(T_u, a) = \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) - \mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z525}} \quad \mathbb{G}_{2,3}^{(c)}(T_u, a) = \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) - 2\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z526}} \quad \mathbb{G}_{2,1}^{(c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z527}} \quad \mathbb{G}_{2,2}^{(c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z528}} \quad \mathbb{G}_{2,3}^{(c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z529}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z530}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) - \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z531}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z532}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{21}$$

$$\boxed{\text{z533}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{21}$$

$$\begin{aligned}
\text{z534} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, b) &= \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{21} \\
\text{z535} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, c) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6} \\
\text{z536} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, c) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\text{z537} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, c) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\text{z538} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_u, 1) &= \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} \\
\text{z539} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_u, 1) &= \frac{-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} \\
\text{z540} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_u, 1) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} \\
\text{z541} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) + 9\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 12\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{84} \\
\text{z542} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - 9\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + 12\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{84} \\
\text{z543} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) + 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + 12\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{84} \\
\text{z544} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{6} \left(\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{6} \\
\text{z545} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6} \\
\text{z546} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z547}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_u, b) = \frac{\sqrt{2} \left(\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z548}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_u, b) = \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z549}} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_u, b) = \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) - 2 \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z550}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z551}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z552}} \quad \mathbb{G}_{2,3}^{(1,1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

Atomic SAMB

- bra: $\langle \frac{1}{2}, \frac{1}{2}; p |, \langle \frac{1}{2}, -\frac{1}{2}; p |, \langle \frac{3}{2}, \frac{3}{2}; p |, \langle \frac{3}{2}, \frac{1}{2}; p |, \langle \frac{3}{2}, -\frac{1}{2}; p |, \langle \frac{3}{2}, -\frac{3}{2}; p |$
- ket: $|\frac{1}{2}, \frac{1}{2}; p\rangle, |\frac{1}{2}, -\frac{1}{2}; p\rangle, |\frac{3}{2}, \frac{3}{2}; p\rangle, |\frac{3}{2}, \frac{1}{2}; p\rangle, |\frac{3}{2}, -\frac{1}{2}; p\rangle, |\frac{3}{2}, -\frac{3}{2}; p\rangle$

$$\boxed{\text{x1}} \quad \mathbb{Q}_0^{(a)}(A_g) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\boxed{\text{x2}} \quad \mathbb{Q}_{2,1}^{(a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$$

$$\boxed{\text{x3}} \quad \mathbb{Q}_{2,2}^{(a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x4}} \quad \mathbb{Q}_{2,1}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{6}i}{12} \\ -\frac{\sqrt{6}i}{12} & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x5}} \quad \mathbb{Q}_{2,2}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{6}}{12} \\ \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{4} & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & \frac{\sqrt{6}}{12} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x6}} \quad \mathbb{Q}_{2,3}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x7}} \quad \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\boxed{\text{x8}} \quad \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x9}} \quad \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & \frac{i}{4} & 0 \\ 0 & 0 & 0 & -\frac{i}{4} & 0 & -\frac{\sqrt{3}i}{12} \\ -\frac{\sqrt{3}i}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & \frac{i}{4} & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ -\frac{i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} \\ 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x10}} \quad \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{12} & 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & -\frac{1}{4} & 0 & \frac{\sqrt{3}}{12} \\ \frac{\sqrt{3}}{12} & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & -\frac{1}{4} & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ -\frac{1}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & \frac{\sqrt{3}}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x11}} \quad \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x12}} \quad \mathbb{Q}_0^{(1,1;a)}(A_g) = \begin{bmatrix} -\frac{\sqrt{3}}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \end{bmatrix}$$

$$\boxed{\text{x13}} \quad \mathbb{G}_{1,1}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & -\frac{i}{4} & 0 \\ 0 & 0 & 0 & \frac{i}{4} & 0 & -\frac{\sqrt{3}i}{4} \\ -\frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 \\ \frac{i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x14}} \quad \mathbb{G}_{1,2}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & -\frac{1}{4} & 0 & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 \\ -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x15}} \quad \mathbb{G}_{1,3}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x16}} \quad \mathbb{M}_{1,1}^{(a)}(T_g) = \begin{bmatrix} 0 & \frac{1}{3} & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{2}}{12} & 0 \\ \frac{1}{3} & 0 & 0 & -\frac{\sqrt{2}}{12} & 0 & \frac{\sqrt{6}}{12} \\ -\frac{\sqrt{6}}{12} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{12} & \frac{\sqrt{3}}{6} & 0 & \frac{1}{3} & 0 \\ \frac{\sqrt{2}}{12} & 0 & 0 & \frac{1}{3} & 0 & \frac{\sqrt{3}}{6} \\ 0 & \frac{\sqrt{6}}{12} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x17}} \quad \mathbb{M}_{1,2}^{(a)}(T_g) = \begin{bmatrix} 0 & -\frac{i}{3} & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{2}i}{12} & 0 \\ \frac{i}{3} & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & -\frac{\sqrt{6}i}{12} \\ \frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{12} & \frac{\sqrt{3}i}{6} & 0 & -\frac{i}{3} & 0 \\ \frac{\sqrt{2}i}{12} & 0 & 0 & \frac{i}{3} & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x18}} \quad \mathbb{M}_{1,3}^{(a)}(T_g) = \begin{bmatrix} \frac{1}{3} & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 & 0 & \frac{\sqrt{2}}{6} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{6} & 0 & 0 & \frac{1}{6} & 0 & 0 \\ 0 & \frac{\sqrt{2}}{6} & 0 & 0 & -\frac{1}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \end{bmatrix}$$

$$\boxed{\text{x19}} \quad \mathbb{M}_3^{(1,-1;a)}(A_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x20}} \quad \mathbb{M}_{1,1}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & -\frac{\sqrt{6}}{18} & \frac{1}{3} & 0 & -\frac{\sqrt{3}}{9} & 0 \\ -\frac{\sqrt{6}}{18} & 0 & 0 & \frac{\sqrt{3}}{9} & 0 & -\frac{1}{3} \\ \frac{1}{3} & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{9} & \frac{\sqrt{2}}{6} & 0 & \frac{\sqrt{6}}{9} & 0 \\ -\frac{\sqrt{3}}{9} & 0 & 0 & \frac{\sqrt{6}}{9} & 0 & \frac{\sqrt{2}}{6} \\ 0 & -\frac{1}{3} & 0 & 0 & \frac{\sqrt{2}}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x21}} \quad \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & \frac{\sqrt{6}i}{18} & \frac{i}{3} & 0 & \frac{\sqrt{3}i}{9} & 0 \\ -\frac{\sqrt{6}i}{18} & 0 & 0 & \frac{\sqrt{3}i}{9} & 0 & \frac{i}{3} \\ -\frac{i}{3} & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{9} & \frac{\sqrt{2}i}{6} & 0 & -\frac{\sqrt{6}i}{9} & 0 \\ -\frac{\sqrt{3}i}{9} & 0 & 0 & \frac{\sqrt{6}i}{9} & 0 & -\frac{\sqrt{2}i}{6} \\ 0 & -\frac{i}{3} & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x22}} \quad \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) = \begin{bmatrix} -\frac{\sqrt{6}}{18} & 0 & 0 & -\frac{2\sqrt{3}}{9} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{18} & 0 & 0 & -\frac{2\sqrt{3}}{9} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ -\frac{2\sqrt{3}}{9} & 0 & 0 & \frac{\sqrt{6}}{18} & 0 & 0 \\ 0 & -\frac{2\sqrt{3}}{9} & 0 & 0 & -\frac{\sqrt{6}}{18} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\boxed{\text{x23}} \quad \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{5}}{4} \\ 0 & 0 & \frac{\sqrt{15}}{20} & 0 & -\frac{3\sqrt{5}}{20} & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{5}}{20} & 0 & \frac{\sqrt{15}}{20} \\ 0 & 0 & -\frac{\sqrt{5}}{4} & 0 & \frac{\sqrt{15}}{20} & 0 \end{bmatrix}$$

$$\boxed{\text{x24}} \quad \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{5}i}{4} \\ 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & \frac{3\sqrt{5}i}{20} & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{5}i}{20} & 0 & -\frac{\sqrt{15}i}{20} \\ 0 & 0 & \frac{\sqrt{5}i}{4} & 0 & \frac{\sqrt{15}i}{20} & 0 \end{bmatrix}$$

$$\boxed{\text{x25}} \quad \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \end{bmatrix}$$

$$\boxed{\text{x26}} \quad \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & 0 & \frac{\sqrt{3}}{4} \\ 0 & 0 & \frac{1}{4} & 0 & -\frac{\sqrt{3}}{4} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & \frac{1}{4} \\ 0 & 0 & \frac{\sqrt{3}}{4} & 0 & \frac{1}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x27}} \quad \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{4} & 0 & -\frac{\sqrt{3}i}{4} \\ 0 & 0 & -\frac{i}{4} & 0 & -\frac{\sqrt{3}i}{4} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & \frac{i}{4} \\ 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & -\frac{i}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x28}} \quad \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x29}} \quad \mathbb{M}_{1,1}^{(1,1;a)}(T_g) = \begin{bmatrix} 0 & \frac{\sqrt{30}}{9} & \frac{\sqrt{5}}{12} & 0 & -\frac{\sqrt{15}}{36} & 0 \\ \frac{\sqrt{30}}{9} & 0 & 0 & \frac{\sqrt{15}}{36} & 0 & -\frac{\sqrt{5}}{12} \\ \frac{\sqrt{5}}{12} & 0 & 0 & -\frac{\sqrt{10}}{30} & 0 & 0 \\ 0 & \frac{\sqrt{15}}{36} & -\frac{\sqrt{10}}{30} & 0 & -\frac{\sqrt{30}}{45} & 0 \\ -\frac{\sqrt{15}}{36} & 0 & 0 & -\frac{\sqrt{30}}{45} & 0 & -\frac{\sqrt{10}}{30} \\ 0 & -\frac{\sqrt{5}}{12} & 0 & 0 & -\frac{\sqrt{10}}{30} & 0 \end{bmatrix}$$

$$\boxed{\text{x30}} \quad \mathbb{M}_{1,2}^{(1,1;a)}(T_g) = \begin{bmatrix} 0 & -\frac{\sqrt{30}i}{9} & \frac{\sqrt{5}i}{12} & 0 & \frac{\sqrt{15}i}{36} & 0 \\ \frac{\sqrt{30}i}{9} & 0 & 0 & \frac{\sqrt{15}i}{36} & 0 & \frac{\sqrt{5}i}{12} \\ -\frac{\sqrt{5}i}{12} & 0 & 0 & \frac{\sqrt{10}i}{30} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{36} & -\frac{\sqrt{10}i}{30} & 0 & \frac{\sqrt{30}i}{45} & 0 \\ -\frac{\sqrt{15}i}{36} & 0 & 0 & -\frac{\sqrt{30}i}{45} & 0 & \frac{\sqrt{10}i}{30} \\ 0 & -\frac{\sqrt{5}i}{12} & 0 & 0 & -\frac{\sqrt{10}i}{30} & 0 \end{bmatrix}$$

$$\boxed{\text{x31}} \quad \mathbb{M}_{1,3}^{(1,1;a)}(T_g) = \begin{bmatrix} \frac{\sqrt{30}}{9} & 0 & 0 & -\frac{\sqrt{15}}{18} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{9} & 0 & 0 & -\frac{\sqrt{15}}{18} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{18} & 0 & 0 & -\frac{\sqrt{30}}{90} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{18} & 0 & 0 & \frac{\sqrt{30}}{90} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} \end{bmatrix}$$

$$\boxed{\text{x32}} \quad \mathbb{T}_{2,1}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x33}} \quad \mathbb{T}_{2,2}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x34}} \quad \mathbb{T}_{2,1}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & 0 & \frac{1}{4} & 0 & \frac{\sqrt{3}}{4} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & -\frac{1}{4} \\ \frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x35}} \quad \mathbb{T}_{2,2}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & 0 & -\frac{i}{4} & 0 & \frac{\sqrt{3}i}{4} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & -\frac{i}{4} \\ \frac{i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x36}} \quad \mathbb{T}_{2,3}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Cluster SAMB

- Site cluster

** Wyckoff: 8b

$$\boxed{\text{y1}} \quad \mathbb{Q}_0^{(s)}(A_g) = \left[\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4} \right]$$

$$\boxed{\text{y2}} \quad \mathbb{Q}_3^{(s)}(A_u) = \left[\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4} \right]$$

$$\boxed{\text{y3}} \quad \mathbb{Q}_{2,1}^{(s)}(T_g) = \left[\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4} \right]$$

$$\boxed{\text{y4}} \quad \mathbb{Q}_{2,2}^{(s)}(T_g) = \left[\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4} \right]$$

$$\boxed{\text{y5}} \quad \mathbb{Q}_{2,3}^{(s)}(T_g) = \left[\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4} \right]$$

$$\boxed{\text{y6}} \quad \mathbb{Q}_{1,1}^{(s)}(T_u) = \left[\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4} \right]$$

$$\boxed{\text{y7}} \quad \mathbb{Q}_{1,2}^{(s)}(T_u) = \left[\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4} \right]$$

$$\boxed{\text{y8}} \quad \mathbb{Q}_{1,3}^{(s)}(T_u) = \left[\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, -\frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4}, \frac{\sqrt{2}}{4} \right]$$

• Bond cluster

** Wyckoff: 12b@12c

$$\boxed{\text{y9}} \quad \mathbb{Q}_0^{(b)}(A_g) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6} \right]$$

$$\boxed{\text{y10}} \quad \mathbb{T}_3^{(b)}(A_u) = \left[\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6} \right]$$

$$\boxed{\text{y11}} \quad \mathbb{Q}_{2,1}^{(b)}(E_g) = \left[-\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556} \right]$$

$$\boxed{\text{y12}} \quad \mathbb{Q}_{2,2}^{(b)}(E_g) = \left[\frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852} \right]$$

$$\boxed{\text{y13}} \quad \mathbb{M}_{2,1}^{(b)}(E_u) = \left[\frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852} \right]$$

$$\boxed{\text{y14}} \quad \mathbb{M}_{2,2}^{(b)}(E_u) = \left[\frac{5\sqrt{2778}i}{1389}, \frac{5\sqrt{2778}i}{1389}, \frac{5\sqrt{2778}i}{1389}, \frac{5\sqrt{2778}i}{1389}, \frac{23\sqrt{2778}i}{5556}, \frac{23\sqrt{2778}i}{5556}, \frac{23\sqrt{2778}i}{5556}, \frac{23\sqrt{2778}i}{5556}, -\frac{43\sqrt{2778}i}{5556}, -\frac{43\sqrt{2778}i}{5556}, -\frac{43\sqrt{2778}i}{5556}, -\frac{43\sqrt{2778}i}{5556} \right]$$

$$\boxed{\text{y15}} \quad \mathbb{M}_{1,1}^{(b)}(T_g) = \left[0, 0, 0, 0, \frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7} \right]$$

$$\boxed{\text{y16}} \quad \mathbb{M}_{1,2}^{(b)}(T_g) = \left[\frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, 0, 0, 0, 0, \frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14} \right]$$

$$\boxed{\text{y17}} \quad \mathbb{M}_{1,3}^{(b)}(T_g) = \left[\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y18}} \quad \mathbb{Q}_{2,1}^{(b)}(T_g) = \left[\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y19}} \quad \mathbb{Q}_{2,2}^{(b)}(T_g) = \left[0, 0, 0, 0, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y20}} \quad \mathbb{Q}_{2,3}^{(b)}(T_g) = \left[0, 0, 0, 0, 0, 0, 0, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2} \right]$$

$$\boxed{\text{y21}} \quad \mathbb{T}_{2,1}^{(b)}(T_g) = \left[0, 0, 0, 0, \frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, -\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14} \right]$$

$$\boxed{\text{y22}} \quad \mathbb{T}_{2,2}^{(b)}(T_g) = \left[-\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, 0, 0, 0, 0, \frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7} \right]$$

$$\boxed{\text{y23}} \quad \mathbb{T}_{2,3}^{(b)}(T_g) = \left[\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, -\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y24}} \quad \mathbb{Q}_{1,1}^{(b)}(T_u) = \left[0, 0, 0, 0, \frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14} \right]$$

$$\boxed{\text{y25}} \quad \mathbb{Q}_{1,2}^{(b)}(T_u) = \left[\frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, 0, 0, 0, 0, \frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7} \right]$$

$$\boxed{\text{y26}} \quad \mathbb{Q}_{1,3}^{(b)}(T_u) = \left[\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y27}} \quad \mathbb{T}_{1,1}^{(b)}(T_u) = \left[\frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}, \frac{i}{2}, 0, 0, 0, 0, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y28}} \quad \mathbb{T}_{1,2}^{(b)}(T_u) = \left[0, 0, 0, 0, \frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}, \frac{i}{2}, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y29}} \quad \mathbb{T}_{1,3}^{(b)}(T_u) = \left[0, 0, 0, 0, 0, 0, 0, \frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}, \frac{i}{2} \right]$$

$$\boxed{\text{y30}} \quad \mathbb{Q}_{3,1}^{(b)}(T_u, 1) = \left[0, 0, 0, 0, \frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7} \right]$$

$$\boxed{\text{y31}} \quad \mathbb{Q}_{3,2}^{(b)}(T_u, 1) = \left[-\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, 0, 0, 0, 0, \frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14} \right]$$

$$\boxed{\text{y32}} \quad \mathbb{Q}_{3,3}^{(b)}(T_u, 1) = \left[\frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, 0, 0, 0, 0 \right]$$

Site and Bond

Table 4: Orbital of each site

#	site	orbital
1	A	$ \frac{1}{2}, \frac{1}{2}; p\rangle, \frac{1}{2}, -\frac{1}{2}; p\rangle, \frac{3}{2}, \frac{3}{2}; p\rangle, \frac{3}{2}, \frac{1}{2}; p\rangle, \frac{3}{2}, -\frac{1}{2}; p\rangle, \frac{3}{2}, -\frac{3}{2}; p\rangle$

Table 5: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	A	A	[1]	[p]	[p]

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 6: 'A' (#1) site cluster (8b), .3.

SL	position (s)	mapping
1	[1.00000, 1.00000, 1.00000]	[1,5,9]
2	[-1.00000,-1.00000, 1.00000]	[2,7,12]
3	[-1.00000, 1.00000,-1.00000]	[3,8,10]
4	[1.00000,-1.00000,-1.00000]	[4,6,11]
5	[-1.00000,-1.00000,-1.00000]	[13,17,21]
6	[1.00000, 1.00000,-1.00000]	[14,19,24]
7	[1.00000,-1.00000, 1.00000]	[15,20,22]
8	[-1.00000, 1.00000, 1.00000]	[16,18,23]

Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 7: 1-th 'A'-'A' [1] (#1) bond cluster (12b@12c), ND, $|\mathbf{v}|=2.0$ (cartesian)

SL	vector ($\mathbf{v}$)	center ($\mathbf{c}$)	mapping	head	tail	$\mathbf{R}$ (primitive)
1	[-2.00000, 0.00000, 0.00000]	[0.00000, 1.00000, 1.00000]	[1,-16]	(8,1)	(1,1)	[0,0,0]
2	[2.00000, 0.00000, 0.00000]	[0.00000,-1.00000, 1.00000]	[2,-15]	(7,1)	(2,1)	[0,0,0]

continued ...

Table 7

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
3	[2.00000, 0.00000, 0.00000]	[0.00000, 1.00000,-1.00000]	[3,-14]	(6,1)	(3,1)	[0,0,0]
4	[-2.00000, 0.00000, 0.00000]	[0.00000,-1.00000,-1.00000]	[4,-13]	(5,1)	(4,1)	[0,0,0]
5	[0.00000,-2.00000, 0.00000]	[1.00000, 0.00000, 1.00000]	[5,-20]	(7,1)	(1,1)	[0,0,0]
6	[0.00000, 2.00000, 0.00000]	[1.00000, 0.00000,-1.00000]	[6,-19]	(6,1)	(4,1)	[0,0,0]
7	[0.00000, 2.00000, 0.00000]	[-1.00000, 0.00000, 1.00000]	[7,-18]	(8,1)	(2,1)	[0,0,0]
8	[0.00000,-2.00000, 0.00000]	[-1.00000, 0.00000,-1.00000]	[8,-17]	(5,1)	(3,1)	[0,0,0]
9	[0.00000, 0.00000,-2.00000]	[1.00000, 1.00000, 0.00000]	[9,-24]	(6,1)	(1,1)	[0,0,0]
10	[0.00000, 0.00000, 2.00000]	[-1.00000, 1.00000, 0.00000]	[10,-23]	(8,1)	(3,1)	[0,0,0]
11	[0.00000, 0.00000, 2.00000]	[1.00000,-1.00000, 0.00000]	[11,-22]	(7,1)	(4,1)	[0,0,0]
12	[0.00000, 0.00000,-2.00000]	[-1.00000,-1.00000, 0.00000]	[12,-21]	(5,1)	(2,1)	[0,0,0]